



Apple buys Indictiv

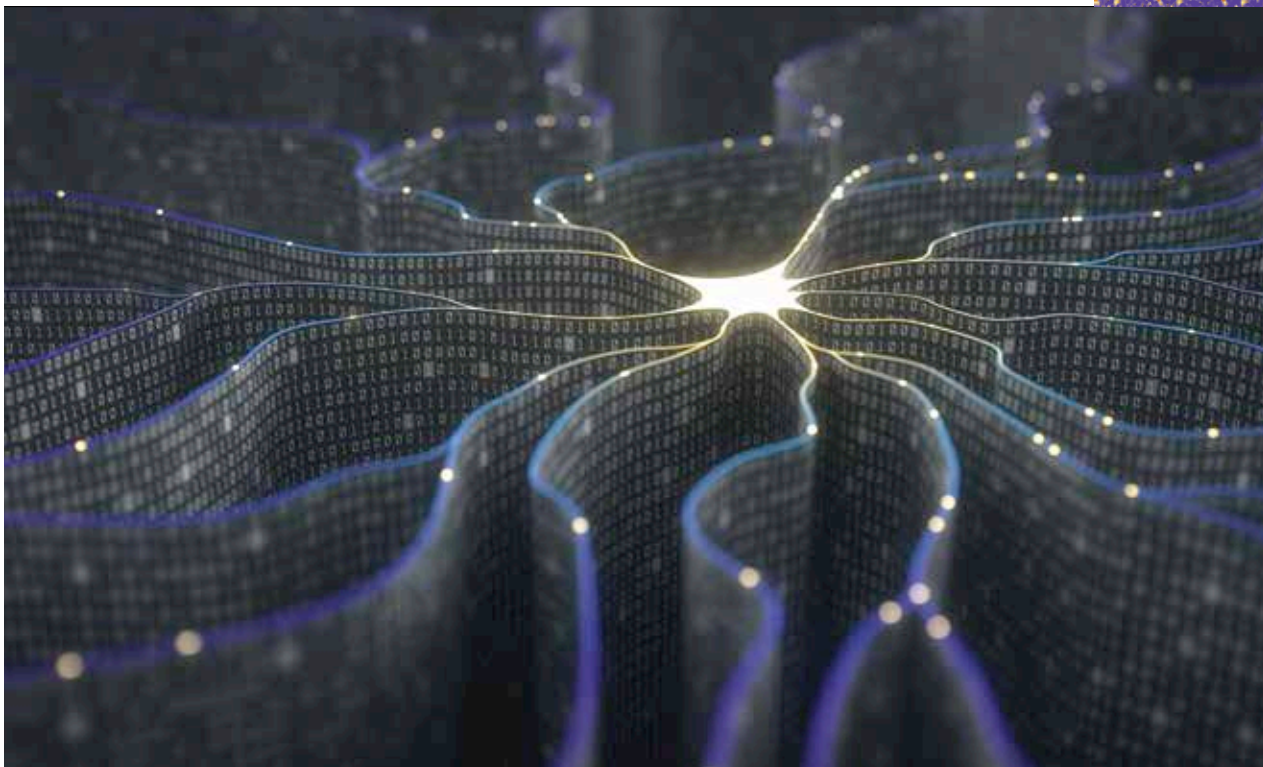
Apple has acquired another AI startup in an effort to clean up data used by Siri. <https://dgit.in/jun20-94>



Microsoft launches Cortana app

MS has launched a dedicated Cortana App on Windows 10, freeing Cortana from the taskbar. <https://dgit.in/jun20-95>

FEATURE



TINY AI

Packing massive computational power into smaller spaces

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I is progressively becoming more intelligent every day, however, it is definitely not getting

greener. In the ceaseless quest to build powerful AI solutions with complex and massive algorithms, enormous amounts of data and computing power is being consumed. This not only has an adverse effect on the environment in terms of carbon emissions, but also reduces the speed and limits of the security of AI applications. As AI is increasingly

becoming more accurate, the burden it poses on the environment proportionally increases as well.

A recent study conducted by researchers at the University of Massachusetts Amherst reveals that training one single algorithm may have five times as much lifetime carbon dioxide emissions as that of an average car. One example is the Turing Natural Language Generation model introduced by Microsoft which is one of the largest AI models ever published at a whopping 17 billion parameters. Such a model, while extremely accurate, is also notoriously power hungry. The relentless search for the highest accuracy possible in AI models has compromised energy efficiency goals. This is where Tiny AI may help. MIT Technology Review in April 2020 named Tiny AI as one of the major technological breakthroughs for this year.

WHAT IS TINY AI AND HOW DOES IT WORK?

Tiny AI is the counter trend to increasing carbon emissions in order to develop more efficient AI. An army of researchers as well as tech giants are developing new algorithms that shrink existing deep-learning models while keeping their capabilities intact. Researchers are attempting to shrink the size of algorithms in AI models, especially those that utilise massive datasets and computational power. One such example is BERT (Bidirectional Encoder Representations from Transformers). BERT is a pre-trained NLP (Natural Language Processing) technique developed by Jacob Devlin and his team at Google. BERT has the ability to understand words and the context in which they are used as well. As a result of this, it can give writing suggestions, finish off sentences, and much more. However, the ability to do this with almost unparalleled accuracy comes at a cost. BERT works with a colossal data set and requires massive amounts of computational power. It has a whopping 340 million data parameters and training it just one time utilises as much electricity as what an average US household would